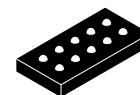


Auto Focus (AF) Controller & Driver

LC898217XC



WLCSP10, 1.04x2.04
CASE 567LF

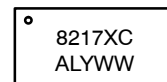
Overview

LC898217XC is an AF control LSI. It consists of 1 system of feedback circuit for AF control.

Features

- Built-in Equalizer Circuit Using Digital Operation
 - ◆ AF Control Equalizer Circuit
 - ◆ Any Coefficient can be Specified by 2-wire Serial I/F (TWIF)
- 2-wire Serial Interface
(The Communication Protocol is Compatible with I²C)
- Built-in A/D Converter
 - ◆ Input 1 Channel
- Built-in D/A Converter
 - ◆ Output 2 Channel (Hall Offset, Constant Current Bias)
- Built-in VGA
 - ◆ Hall Amp
 - ◆ 1 Channel
- Built-in EEPROM
 - ◆ 128 Byte (16 Byte/Page)
- Built-in OSC
- Built-in Constant Current Driver
 - ◆ 110 mA
 - ◆ 1 Channel
- Package
 - ◆ WLCSP 10-pin
- Supply Voltage
 - ◆ V_{DD} (2.6 V to 3.3 V)
- This Device is Pb-Free, Halogen Free/BFR Free and is RoHS Compliant

MARKING DIAGRAM



8217XC = Specific Device Code
 A = Assembly Location
 L = Wafer Lot
 Y = Year
 WW = Work Week

ORDERING INFORMATION

Device	Package	Shipping [†]
LC898217XC-MH	WLCSP10	4000 / Tape & Reel

[†]For information on tape and reel specifications, including part orientation and tape sizes, please refer to our Tape and Reel Packaging Specification Brochure, [BRD8011/D](#).

PIN DESCRIPTION

Table 1. PIN DESCRIPTION

Pin Name	Description
I	Input
P	Power Supply, GND
NC	Not Connect
O	Output
B	Bidirection

- 2-wire serial interface

SCL I 2-wire serial interface clock pin
 SDA B 2-wire serial interface data pin

- Hall interface

BIASO O D/A output (to Hall element)
 OPINP I VGA input (from Hall element)
 OPINM I VGA input (from Hall element)

- Driver interface

OUT1 O Driver output (to Actuator)
 OUT2 O Driver output (to Actuator)

- Power supply pin

V_{DD} P Power Supply
 V_{SS} P GND

- Test pin

PORT B Analog test signal input/output
 Convergence detection monitor output
 VSYNC input

**Process when pins are not used*

PIN TYPE “O” – Ensure that it is set to OPEN.

PIN TYPE “I” – OPEN is inhibited. Ensure that it is connected to the V_{DD} or V_{SS} even when it is unused. (Please contact **onsemi** for more information about selection of V_{DD} or V_{SS}.)

PIN TYPE “B” – If you are unsure about processing method on the pin description of pin layout table, please contact us.

Note that incorrect processing of unused pins may result in defects.

**In case of connecting PORT pin with HOST CPU*

When LC898217XC is power off and HOST CPU is power on, a HOST CPU pin connected with PORT pin have to be fixed “L” level.

LC898217XC

PIN LAYOUT

Table 2. PIN LAYOUT

Circuit Name	Number of PINs	Circuit Name	Number of PINs
Analog	4	Driver	2
Logic	2	Power	2

“PORT” pin has analog function and digital function.

	A	B
1	OUT2	OUT1
2	VSS	VDD
3	PORT	SCL
4	BIASO	SDA
5	OPINM	OPINP

BOTTOM VIEW

Figure 1. Pin Layout

LC898217XC

BLOCK DIAGRAM

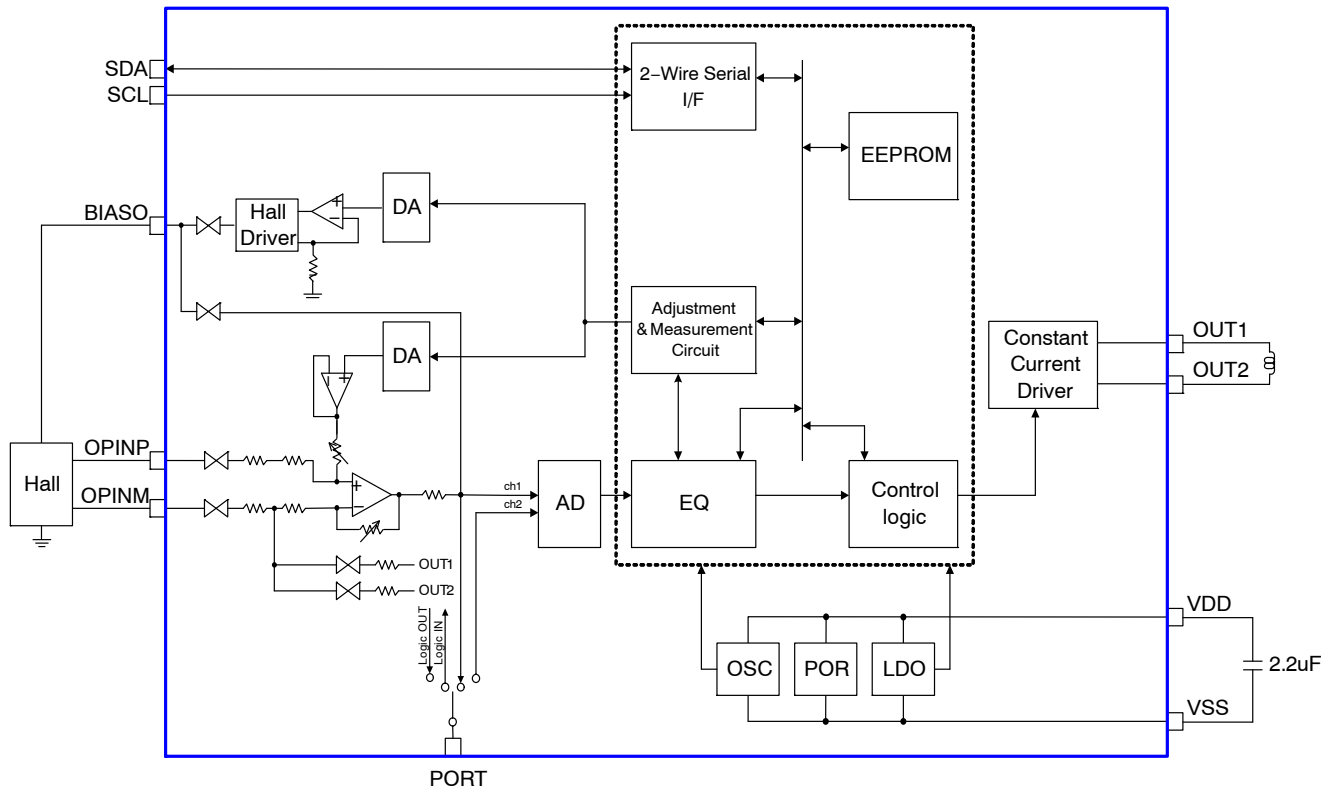


Figure 2. Block Diagram

ELECTRICAL CHARACTERISTICS

Table 3. ABSOLUTE MAXIMUM RATINGS ($V_{SS} = 0\text{ V}$)

Symbol	Item	Condition	Rating	Unit
$V_{DD33\text{ max}}$	Supply voltage	$T_a \leq 25^\circ\text{C}$	-0.3 to 4.6	V
V_{I33}, V_{O33}	Input/output voltage	$T_a \leq 25^\circ\text{C}$	-0.3 to $V_{DD33} + 0.3$	V
T_{stg}	Storage ambient temperature		-55 to 125	$^\circ\text{C}$
T_{opr}	Operating ambient temperature		-30 to 70	$^\circ\text{C}$

Stresses exceeding those listed in the Maximum Ratings table may damage the device. If any of these limits are exceeded, device functionality should not be assumed, damage may occur and reliability may be affected.

Table 4. ALLOWABLE OPERATING RATINGS ($T_a = -30$ to 70°C , $V_{SS} = 0\text{ V}$, 3 V power supply (V_{DD}))

Symbol	Item	Min	Typ	Max	Unit
V_{DD33}	Supply voltage	2.6	2.8	3.3	V
V_{IN}	Input voltage range	0		V_{DD33}	V

Functional operation above the stresses listed in the Recommended Operating Ranges is not implied. Extended exposure to stresses beyond the Recommended Operating Ranges limits may affect device reliability.

Table 5. DC CHARACTERISTICS (Input / output level at $V_{SS} = 0\text{ V}$, $V_{DD} = 2.6$ to 3.6 V , $T_a = -30$ to 70°C)

Symbol	Item	Condition	Min	Typ	Max	Unit	Applicable Pins
V_{IH}	High-level input voltage	CMOS compliant Schmitt	1.4			V	SCL, SDA, PORT
V_{IL}	Low-level input voltage				0.4	V	
V_{OH}	High-level output voltage	$I_{OH} = -2\text{ mA}$	$V_{DD}-0.4$			V	PORT
V_{OL}	Low-level output voltage	$I_{OL} = 2\text{ mA}$			0.4	V	SDA, PORT
R_{dn}	Pulldown resistor		50		220	k Ω	PORT

Product parametric performance is indicated in the Electrical Characteristics for the listed test conditions, unless otherwise noted. Product performance may not be indicated by the Electrical Characteristics if operated under different conditions.

Table 6. DRIVER OUTPUT (OUT1, OUT2) ($V_{SS} = 0\text{ V}$, $V_{DD} = 2.8\text{ V}$, $T_a = 25^\circ\text{C}$)

Symbol	Item	Condition	Min	Typ	Max	Unit	Applicable Pins
I_{full}	Maximum current		105		115	mA	OUT1, OUT2
I_{leak}	Output leak current			1		μA	

Table 7. NON-VOLATILE MEMORY CHARACTERISTICS

Symbol	Item	Condition	Min	Typ	Max	Unit	Applicable Circuit
EN	Endurance				1000	Cycles	EEPROM
RT	Data retention		10			Years	
tWT	Write time				20	ms	

AC CHARACTERISTICS

V_{DD} Supply Timing

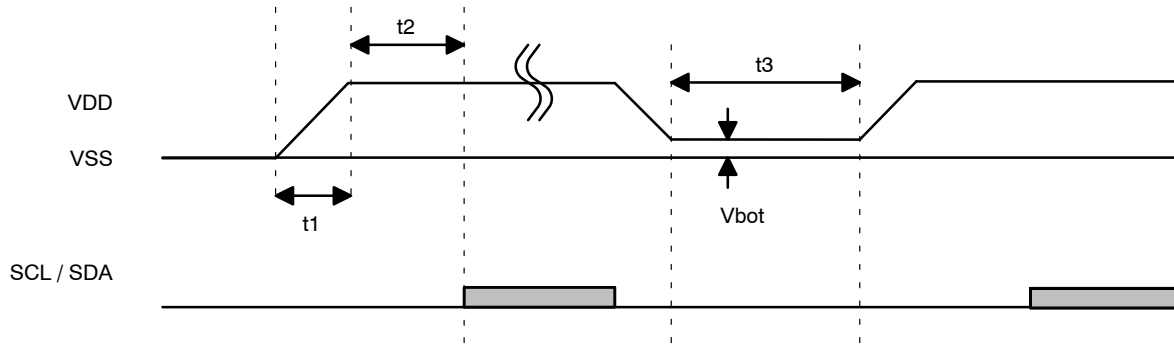


Figure 3. V_{DD} Supply Timing

It is available to use 2-wire serial interface 5 ms later for Power On Reset of V_{DD}.

Table 8. VDD SUPPLY TIMING

Symbol	Item	Min	Typ	Max	Unit
t ₁	V _{DD} turn on time			3	ms
t ₂	2-wire serial interface start time from V _{DD} on	5			ms
t ₃	V _{DD} off time	100			ms
V _{bot}	Bottom Voltage			0.1	V

AC Specification

Figure 4 shows interface timing definition and Table 9 shows electric characteristics.

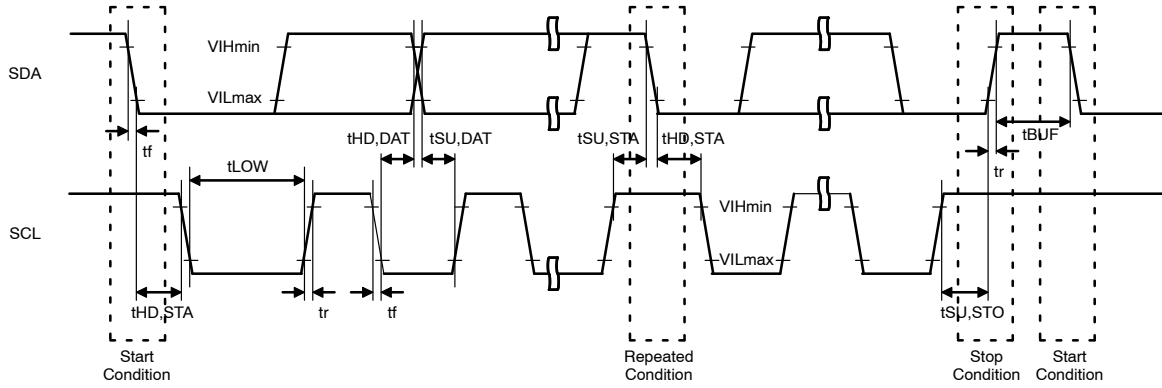


Figure 4. 2-wire Serial Interface Timing Definition

Table 9. ELECTRICAL CHARACTERISTICS FOR 2-WIRE SERIAL INTERFACE (AC CHARACTERISTICS)

Symbol	Item	Pin Name	Fast-mode			Fast-mode Plus			Unit
			Min	Typ	Max	Min	Typ	Max	
F_SCL	SCL clock frequency	SCL			400			1000	kHz
t _{HD,STA}	START condition hold time	SCL SDA	0.6			0.26			μs
t _{LOW}	SCL clock Low period	SCL	1.3			0.5			μs
t _{HIGH}	SCL clock High period	SCL	0.6			0.26			μs
t _{SU,STA}	Setup time for repetition START condition	SCL SDA	0.6			0.26			μs
t _{HD,DAT}	Data hold time	SCL SDA	0 (Note 1)		0.9	0 (Note 1)			μs
t _{SU,DAT}	Data setup time	SCL SDA	100			50			ns
t _r	SDA, SCL rising time	SCL SDA			300			120	ns
t _f	SDA, SCL falling time	SCL SDA			300			120	ns
t _{SU,STO}	STOP condition setup time	SCL SDA	0.6			0.26			μs
t _{BUF}	Bus free time between STOP and START	SCL SDA	1.3			0.5			μs

1. LC898217XC is designed for a condition with typ. 20 ns of hold time. If SDA signal is unstable around falling point of SCL signal, please implement an appropriate treatment on board, such as inserting a resistor.

MECHANICAL CASE OUTLINE

PACKAGE DIMENSIONS

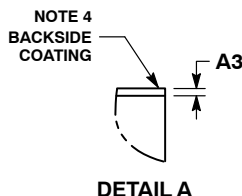
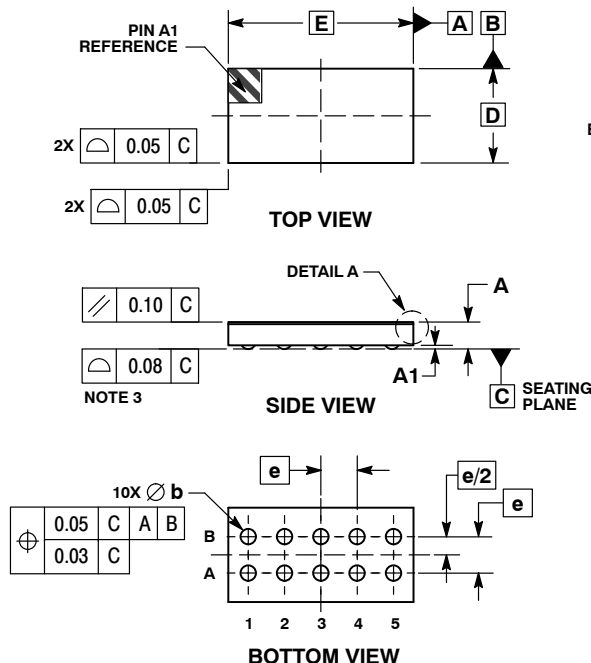
ON Semiconductor®



SCALE 4:1

WLCSP10, 1.04x2.04
CASE 567LF
ISSUE B

DATE 03 JUN 2015

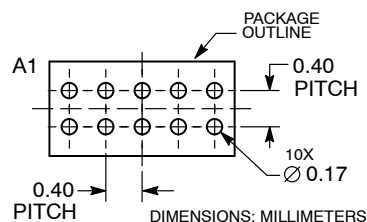


NOTES:

1. DIMENSIONING AND TOLERANCING PER ASME Y14.5M, 1994.
2. CONTROLLING DIMENSION: MILLIMETERS.
3. COPLANARITY APPLIES TO THE SPHERICAL CROWNS OF THE SOLDER BALLS.
4. BACKSIDE COATING IS OPTIONAL.

MILLIMETERS		
DIM	MIN	MAX
A	0.27	0.33
A1	0.04	REF
A3	0.025	REF
b	0.12	0.22
D	1.04	BSC
E	2.04	BSC
e	0.40	BSC

RECOMMENDED SOLDERING FOOTPRINT*



*For additional information on our Pb-Free strategy and soldering details, please download the ON Semiconductor Soldering and Mounting Techniques Reference Manual, SOLDERRM/D.

DOCUMENT NUMBER:	98AON94695F	Electronic versions are uncontrolled except when accessed directly from the Document Repository. Printed versions are uncontrolled except when stamped "CONTROLLED COPY" in red.
DESCRIPTION:	WLCSP10, 1.04X2.04	PAGE 1 OF 1

ON Semiconductor and are trademarks of Semiconductor Components Industries, LLC dba ON Semiconductor or its subsidiaries in the United States and/or other countries. ON Semiconductor reserves the right to make changes without further notice to any products herein. ON Semiconductor makes no warranty, representation or guarantee regarding the suitability of its products for any particular purpose, nor does ON Semiconductor assume any liability arising out of the application or use of any product or circuit, and specifically disclaims any and all liability, including without limitation special, consequential or incidental damages. ON Semiconductor does not convey any license under its patent rights nor the rights of others.

onsemi, **Onsemi**, and other names, marks, and brands are registered and/or common law trademarks of Semiconductor Components Industries, LLC dba "**onsemi**" or its affiliates and/or subsidiaries in the United States and/or other countries. **onsemi** owns the rights to a number of patents, trademarks, copyrights, trade secrets, and other intellectual property. A listing of **onsemi**'s product/patent coverage may be accessed at www.onsemi.com/site/pdf/Patent-Marking.pdf. **onsemi** reserves the right to make changes at any time to any products or information herein, without notice. The information herein is provided "as-is" and **onsemi** makes no warranty, representation or guarantee regarding the accuracy of the information, product features, availability, functionality, or suitability of its products for any particular purpose, nor does **onsemi** assume any liability arising out of the application or use of any product or circuit, and specifically disclaims any and all liability, including without limitation special, consequential or incidental damages. Buyer is responsible for its products and applications using **onsemi** products, including compliance with all laws, regulations and safety requirements or standards, regardless of any support or applications information provided by **onsemi**. "Typical" parameters which may be provided in **onsemi** data sheets and/or specifications can and do vary in different applications and actual performance may vary over time. All operating parameters, including "Typicals" must be validated for each customer application by customer's technical experts. **onsemi** does not convey any license under any of its intellectual property rights nor the rights of others. **onsemi** products are not designed, intended, or authorized for use as a critical component in life support systems or any FDA Class 3 medical devices or medical devices with a same or similar classification in a foreign jurisdiction or any devices intended for implantation in the human body. Should Buyer purchase or use **onsemi** products for any such unintended or unauthorized application, Buyer shall indemnify and hold **onsemi** and its officers, employees, subsidiaries, affiliates, and distributors harmless against all claims, costs, damages, and expenses, and reasonable attorney fees arising out of, directly or indirectly, any claim of personal injury or death associated with such unintended or unauthorized use, even if such claim alleges that **onsemi** was negligent regarding the design or manufacture of the part. **onsemi** is an Equal Opportunity/Affirmative Action Employer. This literature is subject to all applicable copyright laws and is not for resale in any manner.

PUBLICATION ORDERING INFORMATION

LITERATURE FULFILLMENT:

Email Requests to: orderlit@onsemi.com

onsemi Website: www.onsemi.com

TECHNICAL SUPPORT

North American Technical Support:

Voice Mail: 1 800-282-9855 Toll Free USA/Canada

Phone: 011 421 33 790 2910

Europe, Middle East and Africa Technical Support:

Phone: 00421 33 790 2910

For additional information, please contact your local Sales Representative

Mouser Electronics

Authorized Distributor

Click to View Pricing, Inventory, Delivery & Lifecycle Information:

[onsemi:](#)

[LC898217XC-MH](#)